Car sends voice message through satellite

Iridium Communications, Deutsche Telekom IoT, and Toyota have jointly demonstrated standards-based NB-IoT voice messaging over Iridium's low-Earth orbit satellite network. Conducted during Telekom Satellite Day 2026, the trial used a Nordic Semiconductor nRF9151 development board, Deutsche Telekom IoT SIM, and Fraunhofer's AI-powered NESC voice codec. A Toyota vehicle successfully transmitted a voice message via satellite, showcasing continuous connectivity beyond terrestrial mobile coverage. Developed through collaboration in Europe, the technology could support connected vehicles, logistics, agriculture, utilities, and emergency services, providing resilient communication where terrestrial networks are unavailable.



Iridium, Deutsche Telekom IoT, and Toyota demonstrate voice messaging over satellite via Iridium NTN Direct (Credit: https://www.computerweekly.com)

Organic photodiode brings 3.6Gbps to Li-Fi links

Researchers at the University of St Andrews and the University of Cambridge in the UK have developed organic photodiodes capable of supporting 3.6Gbps visible-light communication (VLC) over a one-metre link, marking a major advance for high-speed Li-Fi receivers. The breakthrough utilises a single organic photodiode detector to achieve more than 20 times the previous record rate for an organic receiver. Device engineering addresses bandwidth limits caused by low carrier mobility and parasitic capacitance. Unlike rigid silicon detectors, organic photodiodes can offer flexible, solution-processed, large-area integration, positioning them as viable optical wireless receivers for indoor Li-Fi, IoT nodes, and sustainable wireless sensors.

Efficient computation with fewer qubits enabled

Fujitsu and startup Yaqumo have commenced hardware verification of Fujitsu's STAR quantum computing architecture on operational neutral-atom systems to enable efficient computation with fewer physical qubits. The collaboration evaluates the architecture's compatibility with Yaqumo's ytterbium-based hardware to leverage its native all-to-all qubit connectivity for superior quantum error correction. Concurrently, Fujitsu is adapting its open source Open Quantum Toolchain to manage instruction translation, job scheduling, and device calibration. This integration aims to provide unified, cross-platform cloud access to heterogeneous quantum backends as Yaqumo scales hardware beyond several hundred qubits by fiscal 2027.

AI exoskeleton helps spinal-cord injury patients walk again

Kochi-based startup Astrek Innovations has developed a lower-limb robotic exoskeleton that assists walking movements during rehabilitation. The powered system straps around the hips and legs, utilising motors and machine-learning algorithms trained on six years of healthy movement data to support a patient's gait patterns. Designed for repeated gait training, the patented modular device triggers walking motions, weight-bearing, balance training, and lower-limb movement for individuals recovering from stroke or spinal cord injuries. Field-tested by patients with spinal cord injuries, the technology reduces dependence on upper-body strength compared to traditional callipers, though continuous therapist supervision remains essential to grade and modify the therapy.



It took six years of collecting data on how healthy people sit, stand, and walk to train the algorithms behind the device (Credit: https://thebetterindia.com)

Microwave test system evaluates electronics resilience

The Defence Research and Development Organisation (DRDO) has issued a request for proposal for SHIELD, an S-band high-power microwave integrated evaluation system for lethality and damage. Developed under the 'Technology Development Fund' scheme, the platform evaluates electronic equipment vulnerability under controlled electromagnetic exposure. The system employs gallium nitride (GaN)-based solid-state power amplifiers to generate high-power RF pulses with a peak far-field electric field strength of at least 3kV/m. Engineered to test communication equipment, sensors, and control systems in pulsed mode with flexible duty cycles, SHIELD quantifies electronic degradation and component survivability. Additionally, a drone-mountable configuration enables mobile airborne deployment to simulate diverse directed-energy threat scenarios.